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Review

Recent Advances in Cochlear Implantation

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Abstract: Since the inception of cochlear implantation, the field of technological advancements associated with cochlear implantation has continued to evolve, providing patients with sensorineural hearing loss access with greater sound appreciation capabilities. These advances include evolving cochlear implantation criteria, including increased residual hearing and single-sided deafness; changes in electrode design; options for hearing preservation; and advancements in connectivity, to name a few. This article reviews the various aspects of the recent advancements in relation to cochlear implantation.

Keywords: cochlear implant; otolaryngology; otology; audiology; hearing loss; single-sided deafness

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1. Introduction

In 2020, the World Health Organization estimated that more than 1.5 billion people live with hearing loss, with 430 million experiencing moderate-or-higher levels of hearing loss [1]. Despite improvements in the prevention and early recognition of hearing loss, its prevalence in the United States is projected to nearly double by the year 2060 [2]. If untreated, hearing loss is associated with reductions in cognition, social interaction, employment status, and annual income [3]. For those who qualify, the use of restorative hearing devices, such as cochlear implants, has been associated with significant reductions in cognitive decline and improvements in social interactions and working lives [3,4].

Cochlear implants are surgically placed medical devices that restore sensorineural hearing loss by bypassing damaged portions of the inner ear and directly stimulating the auditory nerve. As of 2019, the U.S. Food and Drug Administration (FDA) reported that there are approximately 736,900 registered cochlear implants worldwide, with around 118,100 devices implanted in adults and 65,000 in children in the United States [5]. Since their approval by the FDA in the 1980s, many advances have been made in relation to cochlear implantations that have led to improvements in patient outcomes. The purpose of this narrative review is to discuss recent advances in cochlear implantation and potential future directions in the field.

2. Expanding Indications

Since cochlear implantation (CI) was first approved by the United States Food and Drug Administration (FDA) for bilateral postlingually deaf adults in 1984, CI candidacy has significantly expanded to patients with less-severe hearing loss, as well as children.

There are three companies with FDA-approved CI devices, including Advanced Bionics®, Cochlear Americas®, and MED-EL®. Each company currently has differing approval for candidacy criteria. Outside of the United States, there are no standardized criteria, with each country's healthcare system establishing their own guidelines regarding CI.

With improvements in CI technology, the indications for CI have also expanded significantly over the years. In 1990, the FDA approved CI for children aged 2 years and older with profound hearing loss [6]. This age was lowered to 12 months in 2000 and 9 months in 2020, as earlier implantation in pediatric patients is associated with more accelerated development in comprehension and expression [7,8]. MED-EL® received approval for implantation in patients over 5 years old with single-sided deafness (SSD) in 2019, while Cochlear Americas ® received a similar approval in 2022 [9]. There are currently numerous studies investigating these devices in patients with preserved degrees of hearing, as well as in children under 9 months [7,10]. From 2005 to 2022, the Centers for Medicare and Medicaid Services (CMS) set the criteria to qualify for CI at a score of $\leq$40% correct on open-set sentence recognition tests in the patient's best aided listening condition, or up to 60% correct if their provider was participating in an approved clinical trial. However, in 2022, the CMS expanded its criteria for CI candidacy to a score of $\leq$60% in all patients, regardless of clinical trial status. With these criteria expansions, it is estimated that up to 2.5 million adults in the US alone will qualify for cochlear implantation, 2.1 million of whom will be 65 years or older [11]. Additionally, with the expanded CMS criteria, a larger percentage of CI users are utilizing a hearing aid in the contralateral ear, giving them the advantage of bimodal hearing.

While the CMS has not yet indicated CI for SSD, many private insurance companies have expanded their qualifying criteria to encompass this population subset. There are several advantages to binaural hearing, including the improved localization of sound and speech recognition in noise, which can only be achieved with true auditory input to both ears [12]. Research has shown that adult patients with SSD experience increased listening fatigue, social isolation, and a decreased quality of life compared to their normal hearing counterparts [13,14]. Cochlear implantation and the use of CIs have been shown to enhance sound localization, tinnitus management, speech recognition in noisy environments, and overall quality of life in adults with SSD [15]. For pediatric patients with SSD, early intervention is critical; research has shown an increase in both speech and language delays and behavioral concerns in children with SSD [16]. Furthermore, recent studies have shown that delayed intervention in children with SSD may lead to "aural preference syndrome", where developing auditory pathways are reorganized towards the better ear [17].

As indications for CI have continued to expand, the way in which patients are referred and assessed for CI candidacy has also continued to evolve. In 2024, the Minimum Speech Test Battery-3 (MSTB-3) was released after the Institute of Cochlear Implant Training (ICIT) gathered a panel of expert CI audiologists to discuss and revise current CI clinical practices [18]. The panel had several key objectives—to provide guidance on when patients should be referred for a CI evaluation and to provide consistency among providers on which test materials should be utilized to assess CI candidacy and CI recipients post-operatively [18]. This allows for a larger population of potential CI candidates to be recommended for appropriate follow-up, as patients have historically been under-referred for candidacy evaluations. This is believed to be due to a lack of knowledge among providers, as well as confusion around referral and candidacy criteria. It also allows clinicians who may not specifically work with cochlear implants on a routine basis (such as primary care providers) to feel empowered in referring their patients for surgical rehabilitative options [19].

Previously, it has been suggested by Zwolan et al. (2020) that a "60/60" referral guideline should be utilized to identify patients who may benefit from a CI evaluation [20]. In this previous study, they found that 95% of patients who met traditional indications for CI had a pure-tone average (PTA) in the better ear that was ≥60 dB HL, while 92% had a better-ear unaided monosyllabic word score that was ≤60% [20]. However, this guideline does not capture potential non-traditional CI patients such as those with asymmetric hearing loss (AHL), SSD, hybrid, or electro-acoustic stimulation (EAS). Therefore, the MSTB-3 recommends placing a greater emphasis on evaluating the ear to be treated; specifically, it recommends that patients be referred for a CI evaluation when they have an unaided PTA of 60 dB HL or greater or an unaided word score of 60% or less in the ear to be implanted rather than in their better ear [18].

Additionally, the MSTB-3 places a much higher emphasis on monosyllabic word tests, such as the consonant–nucleus–consonant (CNC) monosyllabic word test, rather than open-set sentence recognition tests, such as the AzBio sentence tests, to assess CI candidacy and CI recipients post-operatively [18,21,22]. This decision was determined according to recent trends that have been seen across CI clinics in utilizing CNCs following FDA-approved indications for hybrid, EAS, SSD, and AHL patients [18]. The advantages of using CNC word tests to assess CI candidacy and CI recipients include the fact that word tests are less likely to demonstrate post-operative ceiling effects compared to sentence recognition tests and the fact that fewer contextual cues are available when utilizing word stimuli as opposed to sentences [23,24]. The MSTB-3 recommends that open-set sentence recognition tests are only utilized after identifying CI candidates in order to satisfy insurance requirements [18]. It is important to acknowledge, however, that both CNC word lists and AzBio sentences are inherently language-specific and may lead to inaccuracies in non-English-speaking countries or for speakers of diverse dialects, which may lead to inaccuracies. In such cases, locally developed and validated speech test materials are essential to ensure accurate assessments that account for linguistic and cultural variations. Other new recommendations include obtaining unaided audiometric testing within 6 months of an evaluation, completing hearing aid verification prior to speech recognition testing, and utilizing validated questionnaires such as the Cochlear Implant Quality of Life (CIQOL-10), Tinnitus Handicap Inventory (THI), and Speech, Spatial and Qualities of Hearing Scale (SSQ-12) to further evaluate a patient's functional listening abilities pre- and post-operatively [18,25–27]. For patients that are unable to communicate or understand traditional candidacy tests, alternative objective methods, such as pre-operative trans-tympanic electrically evoked auditory brainstem responses and electric middle-latency auditory evoked responses, have proven effective in assessing and confirming candidacy [28,29].

The ever-expanding indications for CI have resulted in the need for the improved tailoring of devices and processing strategies to optimize the device performance for an increasingly diverse patient population.

3. Technological Advances

3.1. Electrode Design

During the last few decades, there have been a number of technological advancements in CI electrode design. Early CIs had a limited number of channels (4–8), which significantly limited the resolution of sound processing and the perception of discrete frequencies [30]. Modern electrode arrays have a greater number of channels (12–24), which has the potential to significantly improve the spatial resolution of the frequency spectrum. Recent studies have demonstrated that more electrodes do not necessarily result in improved hearing outcomes. These studies demonstrate that optimal electrode design involves maximiz-

ing the distribution of channels along the length of the cochlea to prevent cross-channel interaction [31].

Additionally, advances in CI electrode design have allowed for the greater tailoring of the device to the patient's specific needs. There are many considerations relating to electrode design including curvature, length, and flexibility, among others, in order to allow the surgeon to choose devices that may provide the greatest benefit to the patient. Lateral wall (LW), or "straight", electrodes are introduced along the lateral aspect of the scala tympani during insertion, while perimodiolar (PM), or "pre-curved", arrays are designed to lie in closer proximity to the spiral ganglion neurons (SGNs) that are the target of direct electrical stimulation [32]. Of note, cochlear trauma from scalar translocation and tip fold-overs has been found to occur more frequently with perimodiolar arrays than with lateral wall arrays [33].

The length of the electrode is also an important consideration when choosing the appropriate CI for the patient. Traditional CIs were reserved for patients with profound deafness, often requiring longer electrodes to maximize the coverage of the frequency range with electrical stimulation. While each individual may have a variable cochlear duct length (CDL), previously, it has been difficult to estimate the appropriate insertion depth to provide optimal coverage. Recent studies have proposed automated techniques to measure optimal insertion lengths based on computed tomography imaging [34].

While indications for CI are ever-expanding to patients with improved hearing status, the devices have also experienced significant advancements to optimize implantation in these patients. Recent advancements have included additional arrays with greater flexibility to reduce the insertion trauma of viable SGNs, as well as shorter electrodes. The mid-scala (MS) array, which represents a variation in the PM electrode, lies within the mid-portion of the scala tympani to focus stimulation on the higher frequency SGNs and to reduce trauma to distal aspects of the cochlea in order to aid in the preservation of lower-frequency hearing.

Another potential barrier to optimal electrode selection involves the dynamic nature of hearing loss and the decision to choose an electrode for hearing preservation, given that a patient may experience progressive decline. There have been studies regarding the role of genetic testing to provide information regarding the prognostication of a patient's residual hearing status that may help the surgeon determine the best type of electrode for the patient [35].

3.2. Signal Processing

While CI electrodes have evolved since their induction, advancements in signal processing have also resulted in significantly improved hearing outcomes in these patients. Early CIs used analog signal processors (ASPs), which resulted in a limited dynamic range. Modern CIs utilize digital signal processing (DSP), which converts analog signals into digital signals, allowing for greater precision in signal manipulation. Current stimulation strategies intend to improve a CI recipient's ability to perceive distinct frequencies, with the goal of maximizing speech understanding. This is accomplished by limiting the spread of excitation across the SGNs. Early cochlear implants used bipolar and tripolar stimulation methods. However, monopolar stimulation strategies, which are most frequently used today, have proven to result in improved performance for CI recipients while utilizing less energy [36,37].

Additionally, advancements in speech coding strategies have provided greater speech comprehension. Initial cochlear implant strategies were formant-based (F0, F1, F2) and attempted to preserve the fundamental frequency of speech inputs. These strategies delivered few spectral and temporal speech cues [38]. Later strategies, such as SPEAK,

provided additional spectral detail. The development of Continuous Interleaved Sampling (CIS) in the early 1990s represented a major breakthrough in speech perception [39]. The CIS strategy was innovative as it was the first to allow electrodes to be stimulated in sequence. With CIS, temporal cues were better represented and more electrodes were utilized per cycle in comparison to previous strategies. Despite these improvements, CIS lacked temporal fine structure and low-frequency coding, which significantly limited communication in tonal languages, as well as music appreciation [40]. Most modern cochlear implant coding strategies (some of which are proprietary to the manufacturer) are based on CIS and use pulsatile stimulation to improve the preservation of both temporal and spectral speech cues. Although strategies have evolved, their aim to improve speech understanding has remained the same [39,41]. Electric–Acoustic Stimulation (EAS) was developed to improve low-frequency coding through a combination of acoustic and electrical stimulation in individuals with residual acoustic hearing. Studies have demonstrated that EAS users experience improved speech perception amongst background noise, as well as music appreciation, compared to electrical stimulation alone [42]. Further advances encoding low-frequency components with frequency–place matching have improved the experience for users with residual lower-frequency hearing, including patients with SSD [43].

Dynamic range compression algorithms allow for a compression of the significant variability of acoustic signals into the smaller range of electrical signals that the CI electrode may deliver to the SGNs.

Furthermore, advancements in noise reduction (NR) algorithms, along with directional microphones (DMs) and beamforming techniques, enhance the listener's ability to detect the directionality of the signal while attenuating background noise. Single-channel NR techniques strive to filter background signals and allow for enhanced speech signals [44]. Advances in NR algorithms involving machine learning techniques and deep neural networks have resulted in the improved intelligibility of speech for these patients with greater background noise [45]. DMs receive signals from multiple receivers and combine them into a single-channel signal to generate directional patterns to enhance signal perception from certain directions [46]. Advances in adaptive DMs have allowed for greater spatial sound recognition in noisy environments or with moving sound sources [44]. Although there have been significant advancements in sound processing strategies, further studies are needed to continue to improve speech and music appreciation for these patients.

3.3. Connectivity

There have been numerous advancements in CI connectivity that have improved user experience and quality of life. CIs are increasingly equipped with connectivity features that allow users to connect their devices to other external devices such as smartphones and existing assisted-listening devices [47]. Furthermore, advances in teleaudiology practices have allowed CI users to become more independent with their equipment and have decreased the barriers that are traditionally associated with implantation, such as distance to a clinical site for in-person appointments [48].

CI recipients have commonly reported difficulties with speech recognition over the telephone due to a lack of visual cues. With the introduction of Bluetooth Low Energy (BLE) and Made for iPhone® (MFi) in 2014, CI recipients are now able to receive direct wireless audio input from their personal smartphone devices [49]. This allows for substantial improvements in speech recognition over the telephone without impacting the overall battery life of external CI processors [50,51]. With a compatible contralateral hearing aid, bimodal CI patients can receive direct audio input to both ears simultaneously, allowing for a better overall sound quality and performance in speech understanding over the telephone [51].

CI users also commonly experience difficulty with speech recognition in noise and when the speaker is at a distance, even several years post-implantation. Digital audio streaming accessories such as wireless remote microphones and TV streamers are available from all three CI manufacturers, aiming to improve these two areas of difficulty. Wireless remote microphones and TV streamers allow users to receive direct audio input, which gives users a better signal-to-noise ratio (SNR). Research has shown that CI participants perform significantly better with both speech recognition in quiet and in noise when using a remote microphone accessory compared to their CI processor alone [52]. CI recipients also perform better with speech recognition at a distance with the use of a wireless TV streamer compared to their CI processor alone [53].

One of the major patient-related barriers to implantation is access to healthcare services, especially for patients living in rural and remote areas [48]. Recent advances in teleaudiology practices, such as the addition of Remote Check to the Cochlear Nucleus Smart® app, aim to reduce the number of in-person appointments required post-operatively [54,55]. Remote Check allows patients to self-administer audiological threshold and speech recognition tests via wireless streaming. Preliminary research studies have shown a strong reliability between self-administered test results and clinician-driven testing [54]. The use of remote monitoring tools such as Remote Check may reduce the number of unnecessary appointments, as well as the travel time and costs associated with receiving CI-related healthcare services [54,55]. Additionally, all three CI manufacturers offer smartphone-based applications that allow patients to receive remote programming that is more accessible and reliable [56–58].

4. Surgical Techniques

Recent advancements in surgical techniques have shown promising results with improvements in patient outcomes.

4.1. Pre-Operative Simulation Planning

Given the variable anatomy of the human temporal bone, one potential complication of CI surgery is encountering unfavorable visualization or direct access to inner ear structures during the placement of the CI electrode. Recent studies have investigated the three-dimensional (3D) printing of temporal bone models using computed tomography (CT) imaging in order to enhance simulation training, surgical planning, and intra-operative use. These models have been shown to result in improved surgical planning, the avoidance of potential complications, and reduced operating times and costs [59].

4.2. Atraumatic Electrode Insertion Techniques

The atraumatic insertion of CI electrodes has been demonstrated to play a role in hearing preservation and provides an optimal stimulation of auditory neurons [60]. Given the risks of translocation or intracochlear trauma during insertion, there have been numerous considerations for "soft surgery techniques", including smaller, pre-curved, and softer electrodes, as previously discussed, in order to minimize damage to the vulnerable intracochlear structures [61]. While the pure round window (RW) insertion is often considered optimal to reduce intracochlear trauma, extended RW and cochleostomy approaches have also been considered in certain patients to reduce trauma from an unfavorable RW niche [62,63]. Recent studies have also advocated for slower insertion speeds of greater than 25 s that have been shown to reduce intracochlear trauma and prevent basilar membrane rupture and translocation [64]. These mitigating effects of slower insertion times have been postulated to occur through the optimal maintenance of intracochlear fluid pressures during electrode insertion [65].

Robot-assisted insertion techniques have also been demonstrated to improve the consistency of insertion speed and trajectory in CI surgery. Recent studies have demonstrated that robot-assisted electrode array insertion techniques may significantly lower insertion forces and variability compared to manual insertions [66]. Furthermore, there are studies investigating intra-operative navigation and electrophysical feedback to enhance the optimal positioning of CI electrodes during insertion [67,68].

4.3. Cochlear Implantation Under Conscious Sedation

CI surgery has traditionally been performed under general anesthesia without the use of paralysis to monitor facial nerve stimulation intra-operatively. Despite this, there remain many potential medical and cognitive risks associated with general anesthesia, particularly in the elderly population, which represents the majority of CI candidates [69]. Recent studies have demonstrated that CI under conscious sedation has proven to be a safe and effective alternative to general anesthesia and may result in decreased operative times and improved patient satisfaction [70]. Furthermore, the ability to communicate with the patient during the procedure allows the surgeon to evaluate any changes in hearing, balance, or facial nerve function intra-operatively [71]. CI performed under conscious sedation has been demonstrated to have equal outcomes compared to those performed under general anesthesia. There was no significant increase in the rates of complications in cochlear implantation performed under local anesthesia with conscious sedation [70]. Particularly amongst patients with concerns for frailty or cognitive decline, the post-operative course may be improved in patients who undergo CI under sedation versus those who undergo general anesthesia.

5. Biocompatibility and Longevity

Materials and Coatings

Despite advancements in electrode insertion techniques and implant components, the gradual loss of residual hearing over time may occur after implantation. This loss is suspected to be related to a foreign body reaction from the silicone polymer coating, which may result in subsequent inflammation and fibrosis that may impede the electrical current from the electrode to the auditory nerve. There are numerous studies investigating novel "hydrogel" coatings, among others, to reduce this foreign body reaction, decrease resistance on insertion, and prevent potential biofilm formation following CI surgery [72–74].

Additionally, clinicians have advocated for the administration of systemic or local glucocorticoids to reduce the inflammatory effects of CI surgery. Recent studies have investigated the efficacy of glucocorticoids, demonstrating a reduced inflammatory response to electrode insertion trauma with demonstrated improved audiological outcomes [75], although there remains controversy over the efficacy and duration of this benefit [76]. Advanced drug delivery systems, including drug-eluting electrodes, offer a potential approach to protect and enhance residual hearing by eluting anti-inflammatory or growth-promoting agents. These electrodes allow for drug delivery directly within the cochlea, ensuring sustained release to the target area and potentially overcoming the limitations of systemic therapy. The use of drug-eluting electrodes has been associated with reduced impedance, tissue reactions, and inflammatory markers following implantation, as well as improved audiological outcomes, in both human and animal studies [77,78]. While studies have been limited by smaller sample sizes and short follow-up periods, further research holds great promise in better understanding the efficacy of eluting electrodes.

6. Ethical and Social Considerations

Expanding the criteria for CI has significantly increased the number of patients who are candidates for CI surgery. Despite the increasing number of candidates, there remain significant barriers regarding access to appropriate audiological and surgical evaluation. A recent multicenter study reported that only 35.7% of adults who met the criteria for cochlear implantation assessment had a documented discussion about their eligibility for further evaluation, of which only 9.7% were ultimately referred [79]. The study highlighted that patients with advanced age, cognitive disabilities, lower socioeconomic status, and certain races or ethnicities were less frequently informed about their eligibility for cochlear implant candidacy.

Access was further limited by the COVID-19 pandemic. A recent study reported that cochlear implantation was reduced by over 15% among adults aged 65 to 80, and by almost 25% among those 80 years and older [80]. This reduction was estimated to result in more than a three-year setback in the total number of annual cochlear implantations within the United States. Among the pediatric population, patients with prelingual deafness experienced a significant delay between candidacy evaluation and implantation during COVID-19 compared with pre-COVID-19, while there was no significant difference in this interval for patients with post-lingual deafness [81]. Future research may uncover whether these delays in intervention have affected speech and language outcomes among these patients.

7. Conclusions

CI surgery has witnessed significant advancements since the introduction of this invaluable technology in the latter half of the 20th century. This review provides an overview of recent advancements in the field of CI surgery. With the expanding candidacy for implementation in patients with greater residual hearing, there has been an increasing need to improve technology, techniques, and education in order to provide the optimal hearing outcomes for our patients.

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